Integrations Document

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Webhook Integrations Documentation

In this method, the user must provide the URL of an API endpoint, which will be configured in the MindLabs platform. All relevant data received from MindLabs devices will then be forwarded to this URL via HTTP POST requests.

Example Request Body:

JSON Payload

```
{
  "timestamp": 1709316704,
  "signature": "fn2498h293nddwaei92",
  "payload": {
    "type": "packets",
    "data": {
      "id": "AA3630",
      "personalReference": "NEWGO1", // Optional field - shown only if a reference ID is assigned
      "assetName": "Chamber No. 4", // Optional field - visible only when the device is linked to an asset or shipment
      "packets": [
        {
          "timestamp": 1709314784,
          "battery": 100,
          "timeInterval": 60,
          "tempI": 5.3,
          "humidI": 74.9
        },
        {
          "timestamp": 1709315685,
          "battery": 100,
          "timeInterval": 60,
          "tempI": 5.6,
          "humidI": 76.6
        },
        {
          "timestamp": 1709316644,
          "battery": 100,
          "timeInterval": 60,
          "tempI": 5.6,
          "humidI": 73.6,
          "location": {
            "formatted_address": "X42G+GV9, Palaspa, Sakinaka, Chirvat, Maharashtra 410221, India",
            "lng": 73.12768,
            "lat": 18.95136,
            "timestamp": 1709316644
          }
        }
      ]
    }
  }
}
```

API Integrations Documentation

In this method, users are provided with an API key for authorization. Using the API documentation, they can integrate the MindLabs platform into their existing systems. A few example APIs are listed below.

1. Get Devices

After attaching your device to the MindLabs portal, you can retrieve detailed information about all registered devices using the following API request. The response includes configuration details for each device of the specified type.

Request:

cURL Command

```
curl --location 'https://api.mindlabs.cloud/v2/devices?type=ZN_GO&config=true' \
--header 'apikey: API_KEY' \
--data ''
```

Request Queries:

deviceId: ID of the device.

type: Type of the devices you want to fetch.

config: Whether to include configurations of devices in the response.

Response:

Response Example

```
{
  "success": true,
  "data": [
    {
      "shipmentCount": 0,
      "createdAt": 1713188995129,
      "state": "idle",
      "secret": "dc5475d06d7c",
      "sensors": {
        "tempI": "tempI",
        "tempP1": "tempP1",
        "humidI": "humidI"
      },
      "orgId": "Amaris",
      "id": "AA3184",
      "type": "ZN_GO",
      "lastUpdateLocation": {
        "timestamp": 1713221928,
        "lat": 12.72619,
        "lng": 80.02242,
        "formatted_address": "P-43, 8th Cross Street, Mahindra World City, Sengundram, Tamil Nadu 603002, India",
        "battery": 96
      },
      "lastUpdate": {
        "timestamp": 1713237828,
        "fV": "v2.5.3",
```

Response Example (continued)


```
        "tempI": 31.2,
        "tempP1": -301,
        "humidI": 58.5,
        "battery": 95
      },
      "config": {
          "firmwareVersion": "v3.0b-25",
          "loggingInterval": 600,
          "transmissionInterval": 3600,
          "wifiSsid": "mindlabs",
          "wifiPassword": "mindlabs",
          "sendingMode": "active"
      },
      "pendingConfig": null
    },
    {
      "shipmentCount": 12,
      "createdAt": 1711539859349,
      "state": "idle",
      "secret": "dc5475cfba00",
      "sensors": {
        "tempI": "tempI",
        "tempP1": "tempP1",
        "humidI": "humidI"
      },
      "orgId": "Amaris",
      "id": "AA3375",
      "type": "ZN_GO",
      "eventsAgg": null,
      "activeEvents": null,
      "asset": null,
      "lastUpdateLocation": {
        "timestamp": 1712491906,
        "lat": 12.72644,
        "lng": 80.02245,
        "formatted_address": "Mahindra world city 8th Avenue vikasmathra school link
Road Anjur marriage hall, Chennai plant, opposite Force Motors Ltd, Mahindra World
City, Thirutheri R.F., Tamil Nadu 603204, India",
        "battery": 0
      },
      "eventsAggByUtcHour": null,
      "lastUpdate": {
        "timestamp": 1712491906,
        "fV": "v2.5.3",
        "tempI": 31.9,
        "tempP1": -301,
        "humidI": 48.9,
        "battery": 0
      },
      "config": {
          "firmwareVersion": "v3.0b-25",
          "loggingInterval": 600,
          "transmissionInterval": 3600,
          "wifiSsid": "mindlabs",
          "wifiPassword": "mindlabs",
          "sendingMode": "active"
      },
      "pendingConfig": null
    }
  ]
```

Response Example (continued)

```
}
```

Response Description:

1. **shipmentCount:** A numeric value indicating the total number of different shipments the device has been part of.

2. **createdAt:** A 13-digit Unix timestamp representing the exact time when the device was registered on the MindLabs portal.

3. **state:** Describes the status of the device. It can be:

- **idle:** The device is not part of any shipment. This applies to both single-use and multi-use devices.
- **attached:** The device is currently part of an active shipment.
- **deprecated:** Applicable only to single-use devices; indicates the shipment has ended and the device is no longer reusable.

4. **sensors:** Lists the types of sensors integrated into the device. Common sensors include:

- **tempI:** Internal temperature sensor that provides temperature readings in degrees Celsius. This is the default built-in sensor.
- **humidI:** Sensor for measuring relative humidity, displayed in percentage (%).
- **tempP1:** External probe-based temperature sensor, a cylindrical wire that must be explicitly enabled via a config update. Measures temperature in degrees Celsius.

5. **orgId:** The identifier of the organization or company to which the device is assigned.

6. **id:** The unique identifier of the device.

7. **type:** The model/type of the device, such as ZN_GO for Go-type devices or ZN_ANCHOR for Anchor-type devices.

8. **deviceId:** Represents the ID of the device. It serves as a unique identifier and is required when fetching data packets from the device.

9. **lastUpdate:** Contains the most recent data received from the device. This includes sensor readings, battery level, firmware details, and a 10-digit Unix timestamp marking the exact time the data was received.

10. **lastUpdateLocation:** Provides the latest known location of the device. It includes latitude, longitude, battery status, and a human-readable address resolved using Google Maps and other APIs. If the location cannot be resolved, this field may appear undefined. A 10-digit Unix timestamp indicates when the device was at that location.

11. **config:** Reflects the current configuration applied to the device. It may include:

- **loggingInterval:** Data logging interval
- **transmissionInterval:** Data transmission interval
- **wifiSsid:** Wi-Fi SSID
- **wifiPassword:** Wi-Fi Password
- **sendingMode:** Sending mode (active or stock)
- **firmwareVersion:** Firmware version

12. **pendingConfig:** Contains configuration changes that have been scheduled but not yet applied. These updates will be pushed when the device next comes online.

2. Get Device Packets

To retrieve the data packets from a specific device, you can use the following API request. The response will include sensor readings, battery status, and device location within a specified time range.

Request:

cURL Command

```
curl --location 'https://api.mindlabs.cloud/v1/packets?deviceId=GF5A000001&startTime=1
709267145&endTime=1709357400' \
--header 'apikey: API_KEY' \
--data ''
```

Request Queries:

deviceId: Represents the ID of the device. It serves as a unique identifier and is required when fetching data packets from the device.

startTime: A 10-digit Unix timestamp indicating the start time from which you want to fetch data from the device.

endTime: A 10-digit Unix timestamp indicating the end time up to which you want to retrieve data from the device.

Response:

Response Example


```
{
  "success": true,
  "data": [
    {
      "timestamp": 1709314784,
      "battery": 100,
      "timeInterval": 60,
      "tempI": 5.3,
      "humidI": 74.9
    },
    {
      "timestamp": 1709315685,
      "battery": 100,
      "timeInterval": 60,
      "tempI": 5.6,
      "humidI": 76.6
    },
    {
      "timestamp": 1709316644,
      "battery": 100,
      "timeInterval": 60,
      "tempI": 5.6,
      "humidI": 73.6,
      "location": {
        "formatted_address": "X42G+GV9, Palaspa, Sakinaka, Chirvat, Maharashtra
410221, India",
        "lng": 73.12768,
        "lat": 18.95136,
        "timestamp": 1709316644
```

Response Example (continued)

```
      }
    }
  ]
}
```

Response Description:

1. **timestamp:** A 10-digit Unix timestamp, marking the time when the reading was recorded.
2. **sensor readings:** Data from various sensors, such as:

- **templ:** Temperature reading from the internal sensor.
- **humidl:** Humidity reading.
- **battery:** Battery status.

3. **location:** The device's location, represented by latitude, longitude, and a human-readable address. This field may not be present in every object as location data is sent separately and may not appear in every collection cycle.

3. Get Shipment Groups

This API allows you to retrieve shipment groups.

Request:

cURL Command

```
curl --location
'https://api.mindlabs.cloud/v1/shipmentgroups?shipments=true&devices=true' \
--header 'apikey: API_KEY' \
--data ''
```

Request Queries:

shipments: Sends shipments information along with groups

devices: Sends devices information along with shipments and groups

Response:

Response Example


```
{
  "success": true,
  "data": [
    {
      "createdAt": 1697136608249,
      "id": "default",
      "name": "default",
      "type": "shipment",
      "assets": [
        {
          "assetGroupId": "default",
          "assetGroupName": "default",
          "devices": {
            "AG0356": {
              "id": "AG0356",
              "eventsAgg": {},
              "secret": "dc5475cfd640",
```

Response Example (continued)


```
          "shipmentCount": 0,
          "activeEvents": {},
          "lastUpdateLocation": {
            "timestamp": 1713231294,
            "lat": 54.51263,
            "lng": -1.55688,
            "formatted_address": "60 Eden Cres, Darlington DL1 5TW, UK",
            "battery": 52
          },
          "asset": {
            "name": "Hyd to UK",
            "id": "b83dd756-6b55-4754-95e5-50549a9db0d1",
            "assetGroupId": "default",
            "assetGroupName": "default",
            "type": "shipment",
            "startTime": 1706639400,
            "orgId": "demo1",
            "lastUpdate": {
              "timestamp": 1713231294,
              "fV": "v2.2.1",
              "tempI": 8.6,
              "tempP1": -301,
              "humidI": 56.6,
              "battery": 52
            },
            "type": "ZN_GO",
            "aliases": {}
          }
        },
        "AG0131": {
          "id": "AG0131",
          "eventsAgg": {},
          "secret": "dc5475d06ccc",
          "shipmentCount": 0,
          "activeEvents": {},
          "lastUpdateLocation": {
            "timestamp": 1713227513,
            "lat": 54.51262,
            "lng": -1.55688,
            "formatted_address": "60 Eden Cres, Darlington DL1 5TW, UK",
            "battery": 53
          },
          "asset": {
            "name": "Hyd to UK",
            "id": "b83dd756-6b55-4754-95e5-50549a9db0d1",
            "assetGroupId": "default",
            "assetGroupName": "default",
            "type": "shipment",
            "startTime": 1706639400,
            "orgId": "demo1",
            "lastUpdate": {
              "timestamp": 1713227513,
              "fV": "v2.2.1",
              "tempI": 9.3,
              "tempP1": -301,
              "humidI": 53.6,
              "battery": 53
            },
            "type": "ZN_GO",
            "aliases": {}
```

Response Example (continued)


```
            }
          }
        },
        "status": "active",
        "createdAt": 1707064200215,
        "createdBy": "demo1@mindlabs.cloud.",
        "name": "Hyd to UK",
        "expEndTime": 1707150600,
        "sensorSettings": {},
        "startTime": 1706639400,
        "id": "b83dd756-6b55-4754-95e5-50549a9db0d1",
        "type": "shipment",
        "eventsAgg": {}
      }
    ]
  },
  {
    "createdAt": 1695765399556,
    "id": "SG1",
    "createdBy": "support@mindlabs.cloud.",
    "name": "SG1",
    "type": "shipment",
    "assets": []
  }
 ]
}
```


4. Get Loggers Configuration

This API call gets the configuration of multiple loggers based on their IDs.

Authentication Headers:

Request:

cURL Command

```
curl --location
'https://api.mindlabs.cloud/v1/loggers/config?ids[]=AG0207&ids[]=AA1825' \
--header 'apikey: API_KEY' \
--data ''
```

Response:

Response Example


```
{
  "success": true,
  "data": {
    "AA1825": {
      "id": "AA1825",
      "type": "ZN_ANCHOR",
      "config": {
        "firmwareVersion": "v2.5.5_dev_hd",
        "loggingInterval": 300,
        "transmissionInterval": 300,
        "wifiSsid": "zion",
        "wifiPassword": "zion@iot",
        "enableProbe1": false
```

Response Example (continued)

```
      },
      "pendingConfig": null
    },
    "AG0207": {
         ...
        }
  }
}
```

5. Update Loggers Configuration

This API call updates configuration of multiple loggers based on their IDs.

Request

cURL Command

```
curl --location --request PATCH
'https://api.mindlabs.cloud/v1/loggers/config?ids[]=AG0207&ids[]=AA1825' \
--header 'Content-Type: application/json' \
--header 'apikey: API_KEY' \
--data '{
    "config": {
        "firmwareVersion": "v2.5.5_dev_hd",
        "loggingInterval": 300,
        "transmissionInterval": 300,
        "wifiSsid": "zion",
        "wifiPassword": "zion@iot",
        "enableProbe1": false
    }
}'
```

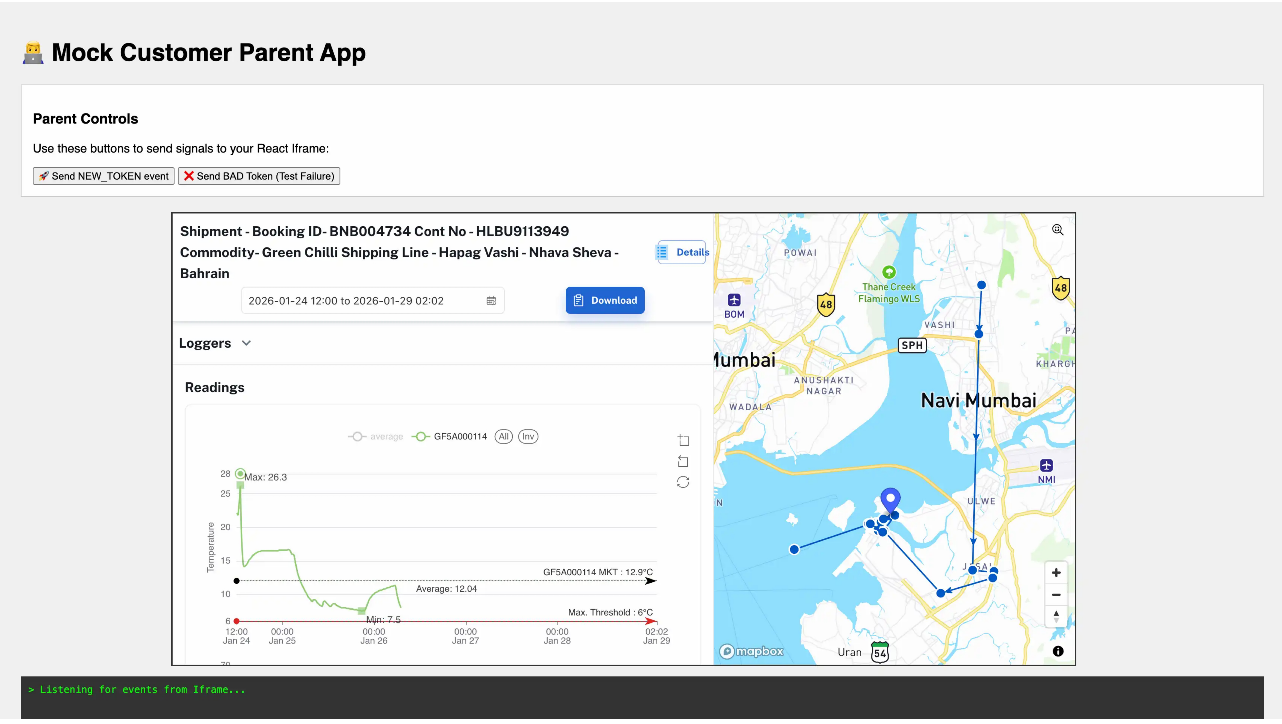
Response:

Response Example

```
{
  "success": true
}
```

Inline Frame Integration Documentation

An inline frame, or <iframe>, is an HTML element used to embed an external webpage or document within the current page. Mindlabs application can be integrated into other web applications using an iframe. This allows users to access Mindlabs features without leaving the host application.



To integrate Mindlabs app using an iframe, you can use the following HTML code snippet:

html

```html
<iframe
  id="myIframe"
  src="https://app.mindlabs.cloud/track/GF5A000001?iframeOrgId=YOUR_ORG_ID&iframeToken=IFRAME_TOKEN"
/>
```

IFRAME_TOKENs can be generated using the following API:

curl

```
curl --location 'https://api.mindlabs.cloud/v1/auth/generate-iframe-token' \
--header 'apikey: API_KEY' \
--data ''
```

If a token expires while an iframe is in use, iframe sends a message to the parent window indicating that the token has expired. The parent window can then handle this message by generating a new token and passing it to the iframe to continue functioning without interruption.

Message format from iframe to parent window:

json

```json
{ type: 'TOKEN_EXPIRED' }
```

Message format from parent window to iframe:

json

```json
{ type: 'NEW_TOKEN', iframeToken: FRESH_TOKEN, iframeOrgId: YOUR_ORG_ID }
```

Example of handling token expiration in the parent window:

javascript

```javascript
window.addEventListener('message', (event) => {
  if (event.data.type === 'TOKEN_EXPIRED') {
    // Generate a new iframe token
    const newToken = generateNewIframeToken(); // Implement this function to get a new
token

    // Send the new token back to the iframe
    const iframe = document.getElementById('myIframe');
    iframe.contentWindow.postMessage({
      type: 'NEW_TOKEN',
      iframeToken: newToken,
      iframeOrgId: 'YOUR_ORG_ID'
    }, '*');
  }
});
```